

J. Manoj
192572143

SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

SOFTWARE: CISCO PACKET TRACER, WIRESHARK

EXPERIMENTS

QUESTIONS:4

Total Marks:40

1. Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.
2. Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.
3. Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.
4. Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

81/100

5. Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

	Understanding of Concepts (25%)	Experimental Design (30%)	Use of Tools (30%)	Analysis of Results (15%)	Documentation (10%)
Q1	2	2.5	2	1	1
Q2	2	2	1.5	1	.5
Q3	2.5	3	2	1.5	.5
Q4	1.5	2	1.5	1	1
Q5	2	2.5	1	1.5	1

8.5
7.5
9.5
7
8

40.5

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology.
(BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Experiments

Code of Conduct:

I __j.manoj(192572143)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

j.manoj

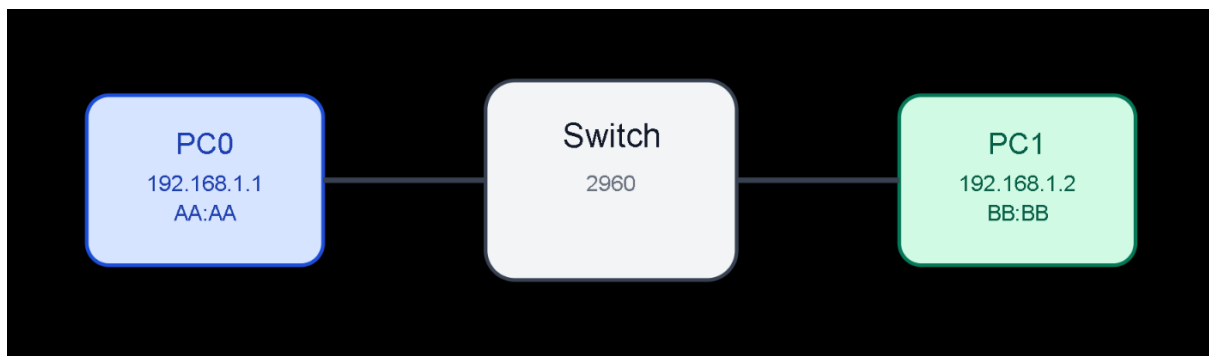
QUESTION 1:

Design a simple LAN consisting of two PCs connected through a switch using Cisco Packet Tracer. Configure appropriate IP addresses for both hosts and verify connectivity using the ping command. Simultaneously capture the packet exchange using Wireshark. Identify the Ethernet frame fields such as Source MAC Address, Destination MAC Address, EtherType, and Frame Check Sequence (FCS). Explain how the Data Link layer uses MAC addressing to ensure successful frame delivery within the local network.

Aim:

To design a simple Local Area Network (LAN) using two PCs connected through a switch in Cisco Packet Tracer, configure IP addresses, verify communication using the ping command, and capture Ethernet frames in Wireshark for analysis.

Network Diagram:



IP Address Configuration Table:

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.1	255.255.255.0
PC1	FastEthernet0	192.168.1.2	255.255.255.0
Switch	VLAN 1	Not Required	—

Procedure:

1. Open Cisco Packet Tracer.
2. Drag 2 PCs and 1 Switch (2960).
3. Connect both PCs using Copper Straight-Through Cable.
4. Configure IP addresses.
5. Open Command Prompt on PC0.
6. Execute: ping 192.168.1.2
7. Start Wireshark and capture packets.
8. Stop capture after successful ping.

Ping Output:

PC0> ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

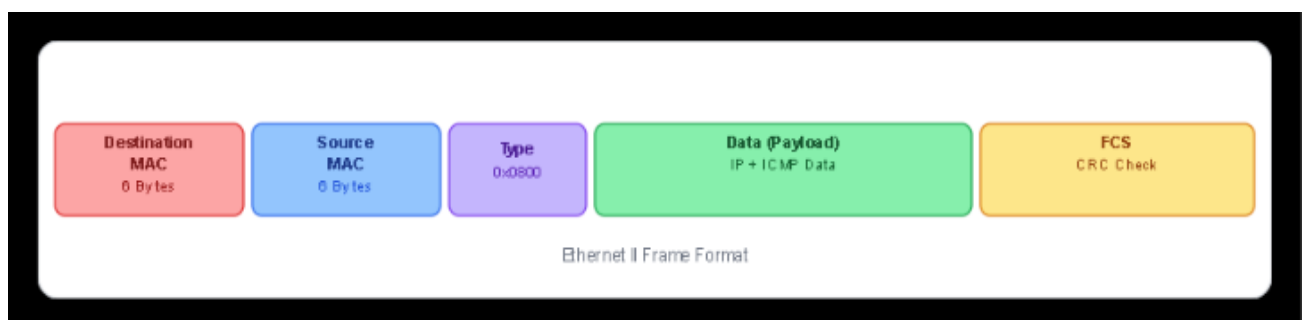
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

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Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Packets: Sent = 4, Received = 4, Lost = 0

Ethernet Frame Structure:



Ethernet Frame Analysis Table:

Field	Size	Example Value	Purpose
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Destination MAC	6 Bytes	BB:BB:BB:BB:02	Identifies receiving device
Source MAC	6 Bytes	AA:AA:AA:AA:01	Identifies sender
EtherType	2 Bytes	0x0800	Indicates IPv4 packet
Payload	46–1500 Bytes	ICMP Data	Carries upper-layer data
FCS	4 Bytes	CRC Value	Error detection

Working Principle:

The Data Link Layer (Layer 2) uses MAC addresses for communication inside a LAN. When PC0 sends data, it places the destination MAC address inside the Ethernet frame. The switch reads this address, checks its MAC table, and forwards the frame only to PC1. The Frame Check Sequence (FCS) verifies that the frame was not corrupted during transmission.

Result:

A LAN was successfully created. Ping communication was successful, and Ethernet frame fields were identified using Wireshark.

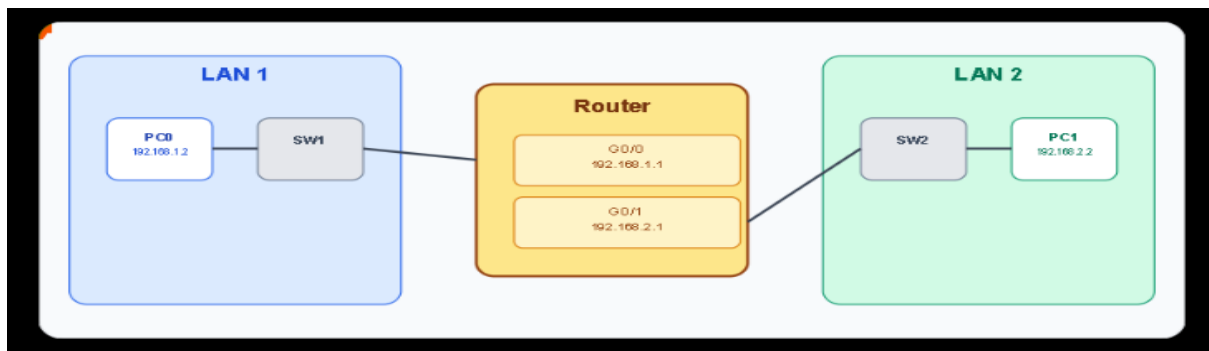
QUESTION 2:

Create a network topology consisting of two LANs connected through a router in Cisco Packet Tracer. Configure IPv4 addressing, subnet masks, and default gateways for all devices, and verify end-to-end communication using the ping command. Capture the ICMP packets using Wireshark and analyze the IPv4 header fields, including Source IP Address, Destination IP Address, Time-To-Live (TTL), Protocol Number, and Header Length. Explain the role of the Network layer in forwarding packets between different networks.

Aim:

To connect two different LANs using a router, configure IPv4 addresses and default gateways, verify end-to-end communication, and analyze IPv4 header fields.

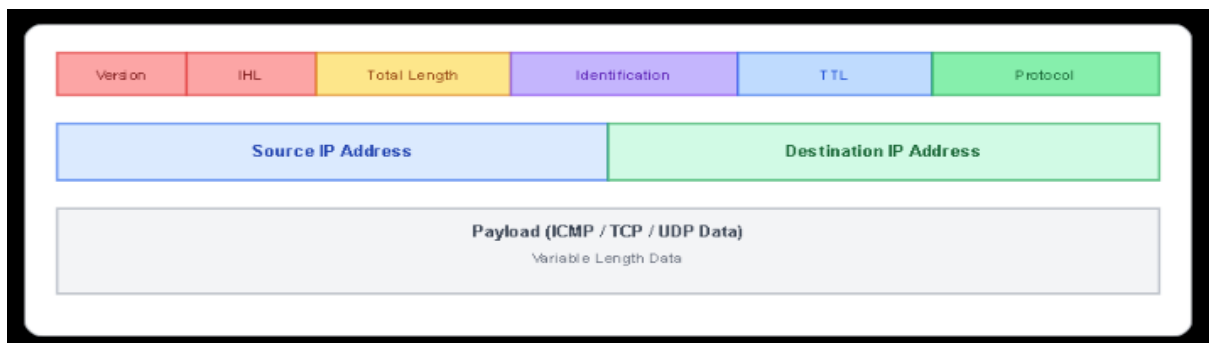
Network Topology:



Address Configuration:

Device	IP Address	Subnet Mask	Gateway
PC0	192.168.1.2	255.255.255.0	192.168.1.1
Router G0/0	192.168.1.1	255.255.255.0	—
Router G0/1	192.168.2.1	255.255.255.0	—
PC1	192.168.2.2	255.255.255.0	192.168.2.1

IPv4 Header Format:



IPv4 Header Analysis:

Header Field	Observed Value	Description
Source IP	192.168.1.2	Sender address
Destination IP	192.168.2.2	Receiver address
TTL	128	Prevents routing loops
Protocol	1	ICMP

Header Length	20 Bytes	Standard IPv4 Header
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Packet Forwarding Process:



Network Layer Explanation:

The Network Layer (Layer 3) is responsible for logical addressing and routing. Unlike switches, routers use IP addresses to determine the destination network. Every time a packet passes through a router, the TTL value decreases by one, preventing infinite routing loops.

Result:

Two different LANs successfully communicated through the router, and IPv4 header fields were analyzed.

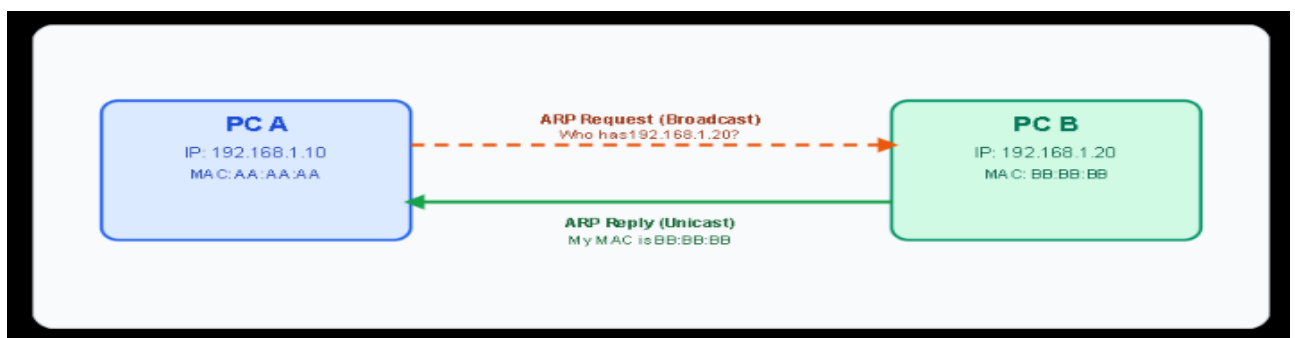
QUESTION 3:

Construct a local area network with multiple hosts connected through a switch in Cisco Packet Tracer. Clear the ARP cache on one of the hosts and initiate communication with another host using the ping command. Capture the communication in Wireshark and identify the ARP Request and ARP Reply packets. Analyze the sender and target MAC/IP addresses and explain how the Address Resolution Protocol resolves logical IP addresses into physical MAC addresses before data transmission.

Aim:

To observe ARP Request and ARP Reply messages by clearing the ARP cache and initiating communication between two hosts.

ARP Working Diagram:



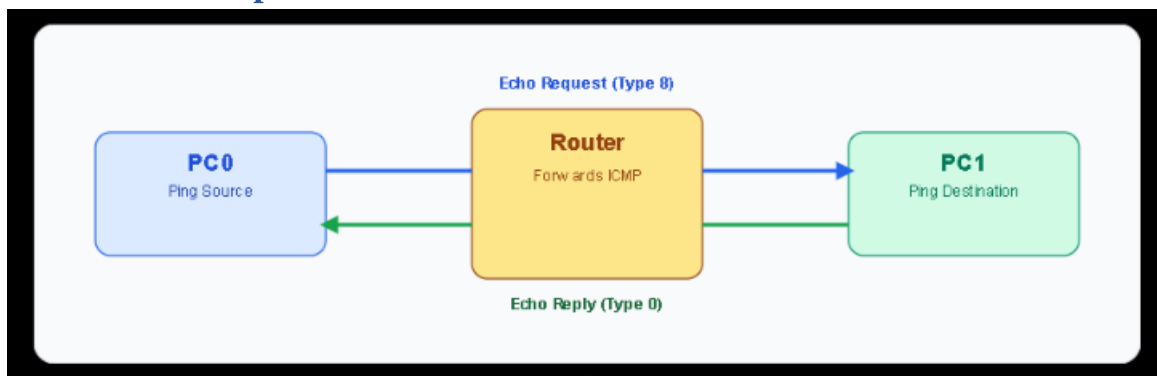
ARP Packet Analysis:

Field	ARP Request	ARP Reply
Sender IP	192.168.1.10	192.168.1.20
Sender MAC	AA:AA:AA	BB:BB:BB
Target IP	192.168.1.20	192.168.1.10
Target MAC	00:00:00:00	AA:AA:AA
Transmission	Broadcast	Unicast

ARP Resolution Flow:

1. PC A checks its ARP cache.
2. MAC address not found.
3. Broadcast ARP Request.
4. PC B receives the request.
5. PC B sends ARP Reply.
6. PC A stores the MAC address.
7. Data transmission begins.

ARCacheExample:



IP Address	MAC Address	Status
192.168.1.20	BB:BB:BB	Dynamic
192.168.1.1	CC:CC:CC	Dynamic

Explanation:

ARP acts as a bridge between Layer 3 (IP) and Layer 2 (MAC). Before any packet is transmitted, the sender must know the receiver's MAC address. ARP automatically performs this resolution, making Ethernet communication possible.

Result:

ARP Request and ARP Reply packets were successfully captured and analyzed.

Q4. ICMP Echo Request, Echo Reply & RTT

Course Outcome: Analyze ICMP packets and network troubleshooting.

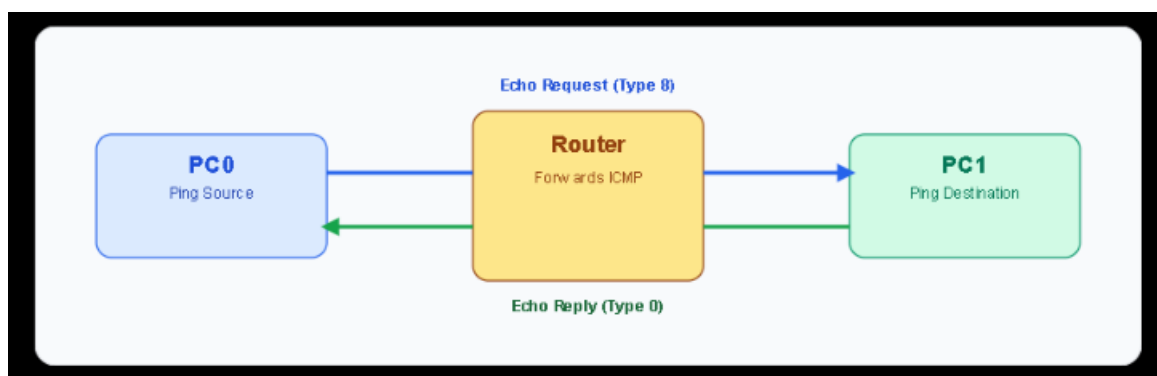
QUESTION 4:

Develop a simple network consisting of two PCs connected through a router in Cisco Packet Tracer. Configure the required IP addresses and test connectivity using the ping command. Capture the ICMP traffic using Wireshark and identify the Echo Request and Echo Reply messages. Examine important ICMP fields such as Type, Code, Identifier, and Sequence Number, and calculate the Round Trip Time (RTT). Discuss how ICMP assists in network troubleshooting and connectivity verification.

Aim:

To configure two PCs connected through a router, capture ICMP packets, identify Echo Request and Echo Reply, and calculate Round Trip Time.

ICMP Communication Diagram:



ICMP Header Fields:

Field	Value	Purpose
Type	8 / 0	Request / Reply
Code	0	Normal ICMP
Identifier	1024	Matches request & reply
Sequence Number	1	Packet order
Checksum	Valid	Error detection

RTT Calculation Table:

Packet	Time (s)
Echo Request Sent	10.500
Echo Reply Received	10.502
RTT	0.002 s (2 ms)

Formula:

$$RTT = T_{reply} - T_{request}$$

$$RTT = 10.502 - 10.500 = 0.002 \text{ seconds}$$

Importance of ICMP:

Function	Description
Ping	Tests connectivity
RTT	Measures latency
Traceroute	Finds routing path
Error Reporting	Destination unreachable
Diagnostics	Network troubleshooting

Result:

ICMP Echo Request and Echo Reply were successfully analyzed, and RTT was calculated as 2 ms.

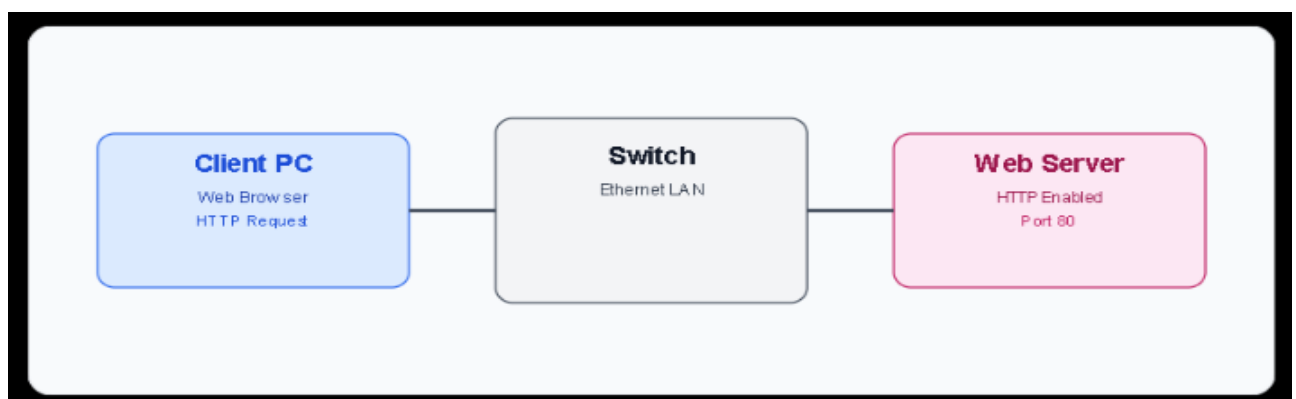
QUESTION 5:

Create a client-server network in Cisco Packet Tracer by configuring a web server and a client PC. Enable the HTTP service on the server and access the hosted webpage from the client using a web browser. Capture the communication in Wireshark and identify the TCP three-way handshake, followed by the HTTP GET request and HTTP response messages. Analyze the application-layer communication and explain how TCP provides reliable delivery for HTTP traffic through connection establishment, acknowledgments, and ordered data transfer.

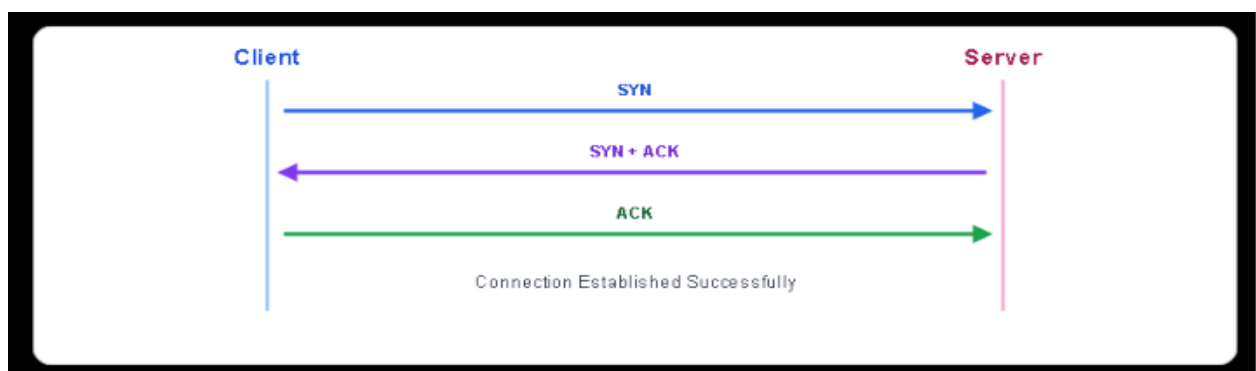
Aim:

To configure a web server and client in Cisco Packet Tracer, enable HTTP service, access the webpage, and analyze TCP three-way handshake along with HTTP GET and Response messages.

Client–Server Architecture:



TCP Three-Way Handshake:



HTTP Communication Flow:



HTTP Messages:

Message	Purpose
SYN	Start TCP connection
SYN-ACK	Server accepts request
ACK	Connection established
HTTP GET	Client requests webpage
HTTP 200 OK	Server sends webpage

TCP Reliability Features:

Feature	Explanation
Connection Establishment	Uses three-way handshake
Sequence Number	Maintains correct order
ACK	Confirms successful delivery
Retransmission	Resends lost packets
Flow Control	Prevents receiver overload

Application Layer Explanation:

HTTP operates at the Application Layer, while TCP operates at the Transport Layer. Before any webpage is transferred, TCP establishes a reliable connection using SYN, SYN-ACK, and ACK. After the connection is established, the client sends an HTTP GET request, and the server replies with an HTTP Response containing the webpage. TCP guarantees reliable, ordered, and error-free delivery.

Result:

The client successfully accessed the web server. TCP handshake and HTTP communication were captured and analyzed using Wireshark.